

AI Labs : AI Solutions - Ank Kumar

EAAP Enterprise AI Automation Platform

Amazon Bedrock Knowledge Base : BIAW4U4TIL · ap-southeast-2
Answer from documents : Claude Haiku 4.5 · au.anthropic.claude-haiku-4-5-20251001-v1:0
Design with the framework : Claude Sonnet 4.5 · au.anthropic.claude-sonnet-4-5-20250929-v1:0

QUERY

Answer from documents

Design with the framework

Answers only from your documents. Every claim is cited to a retrieved passage.

What are the core stages of the Enterprise AI Automation Framework?

Run query

New conversation

first text 4477 ms · total 6964 ms

Start from a known question

What are the core stages of the Enterprise AI Automation Framework?

How does AAIT implement AI-augmented incident triage?

What governance controls sit between an AI decision and an automated action?

DESIGN Propose an AI Solution architecture using the Enterprise AI Automation Framework.

RESPONSE

Core Stages of the Enterprise AI Automation Framework

The EAAF comprises **nine sequential stages** 2 4:

Input & Signal Processing

1. **OBSERVE** – Capture signals and context from enterprise sources 2 4

2. **CORRELATE** – Correlate, enrich, and analyze signals to extract meaningful patterns 2 4

3. **QUALIFY / TRIGGER** – Evaluate whether processed signals warrant action 2 4

Data Preparation & State Management

4. **INGEST & ORCHESTRATE** – Normalize, enrich, route signals, and apply workflow management 2 4

5. **PERSIST STATE** – Store transaction state, context, and audit trails for compliance 2 4

Decision & Governance

6. **AI DECISION / REASONING** – Execute core AI processing with sub-chain: Retrieval → Policy → Prompt → LLM/AI Reasoning → Structured Decision 2 5

7. **GOVERN / DECIDE** – Apply confidence thresholds, policy constraints, and human oversight; branches into automated or human action paths 2 4 5

Execution & Learning

8. **EXECUTE** – Implement decision via enterprise systems/APIs 2 4

9. **OUTCOME** – Capture results, feedback, and evidence; connects to offline analytics for continuous improvement 2 4

Feedback Loop: Stage 9 connects to offline analytics and continuous solution improvement, which feeds back to earlier stages 1 2.

SOURCES

Enterprise_AI_Automation_Framework.pdf https://ank-aws-learning-144160833556.s3.ap-southeast-2.amazonaws.com/ey-fde/Enterprise_AI_Automation_Framework.pdf

Enterprise_AI_Automation_Framework.docx https://ank-aws-learning-144160833556.s3.ap-southeast-2.amazonaws.com/ey-fde/Enterprise_AI_Automation_Framework.docx

TRACE

USER QUERY 67 chars	KNOWLEDGE ROUTING grounded · top_k 5	BEDROCK KNOWLEDGE BASE Retrieve 2606 ms	RETRIEVED CHUNKS 5 passages	CLAUDE ConverseStream 4357 ms	GROUNDED RESPONSE end_turn
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MODE grounded	TOTAL LATENCY 6964 ms	TIME TO FIRST TOKEN 1871 ms
RETRIEVAL 2606 ms	GENERATION 4357 ms	RESPONSE STATUS 200 OK

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KNOWLEDGE BASE ID BIAW4U4TIL	RETRIEVAL OPERATION bedrock-agent-runtime:Retrieve top_k=5	MODEL au.anthropic.claude-haiku-4-5-20251001-v1:0
TOKENS IN / OUT 1833 / 349		
RETRIEVED CHUNKS		
[1] Enterprise_AI_Automation_Framework.pdf0.5245 https://ank-aws-learning-144160833556.s3.ap-southeast-2.amazonaws.com/ey-fde/Enterprise_AI_Automation_Framework.pdf Data Accumulation: Information collected at each stage (signals, knowledge, prompts, reasoning outputs) builds upon and informs subsequent stages. Stage 6 Decision Point: Stage 6 (Confidence/Governance) branches into two parallel paths - "Automated Path" and "Human Approval" - both converging at Stage 7, indicating conditional logic based on confidence levels and governance rules. Feedback Loop: Stage 8 (Offline Improvement) connects back to earlier stages conceptually, suggesting that analysis and simulation results inform future iterations of the model and knowledge base. Input Sources: Stages 1-3 represent input preparation (ingestion, retrieval, construction). Processing Stage: Stage 4 represents the core AI reasoning and analytical processing. Output Stage: Stage 5 generates the structured decision output. Control Stage: Stage 6 applies governance controls and determines execution path. Implementation Stage: Stage 7 executes the decision and captures outcomes. Learning Stage: Stage 8 continuously improves the system based		
[2] Enterprise_AI_Automation_Framework.pdf0.5099 https://ank-aws-learning-144160833556.s3.ap-southeast-2.amazonaws.com/ey-fde/Enterprise_AI_Automation_Framework.pdf EXECUTE flows to stage 9 (OUTCOME) 8. OUTCOME leads to OFFLINE ANALYTICS/GOVERNANCE 9. OFFLINE ANALYTICS/GOVERNANCE connects to CONTINUOUS SOLUTION IMPROVEMENT 10. CONTINUOUS SOLUTION IMPROVEMENT appears to create a feedback loop back to earlier stages 11. All major stage transitions are indicated by downward-pointing triangular arrows </relationships> <description> This is a comprehensive process flow diagram depicting the Enterprise AI Automation Framework (EAAF), illustrating a nine-stage end-to-end workflow for enterprise signal processing and AI-driven decision making. The diagram flows vertically from top to bottom, beginning with enterprise signals as the entry point. The framework progresses through the following sequential stages: Stage 1 (OBSERVE): The entry point captures signals and contextual information from enterprise sources. Stage 2 (CORRELATE): Received signals undergo correlation, enrichment, and analysis to derive meaningful insights. Stage 3 (QUALIFY/TRIGGER): The system evaluates whether		
[3] Enterprise_AI_Automation_Framework.pdf0.4956 https://ank-aws-learning-144160833556.s3.ap-southeast-2.amazonaws.com/ey-fde/Enterprise_AI_Automation_Framework.pdf </relationships> <description> This is a comprehensive process flow diagram illustrating the BMRS (Battlefield Resource Management System) instantiation of the Enterprise AI Automation Framework (EAAF) signal-to-action pattern. The diagram presents an 8-stage sequential workflow designed to automate intelligent decision-making for munitions replenishment and logistics operations in military contexts. The workflow begins with Stage 1 (Domain Signal Ingestion), where five types of military operational signals are collected: battlefield consumption rates, current inventory levels,		

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engagement data, intelligence information, and logistics signals. These raw inputs feed into Stage 2 (Domain Knowledge Retrieval), which aggregates domain-specific knowledge including munitions catalog information, military doctrine, historical consumption patterns, supply network data, and authorization rules. Stage 3 (Prompt Construction) synthesizes the ingested signals and retrieved knowledge into a structured prompt for AI processing, incorporating mission context, current

[4]Enterprise_AI_Automation_Framework.pdf0.4037

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<analysis> <image_type> Process flow diagram / Data flow diagram with hierarchical structure </image_type> <title> End-to-End Enterprise AI Automation Framework (EAAF) Flow - From Enterprise Signals to Continuous Solution Improvement </title> <caption> Complete workflow diagram illustrating the nine-stage process of the Enterprise AI Automation Framework, from initial signal observation through continuous solution improvement, including decision points for automated versus human action. </caption> <extraction> 1. ENTERPRISE SIGNALS (input source) 2. OBSERVE - Capture signals and context 3. CORRELATE - Correlate, enrich and analyze 4. QUALIFY / TRIGGER - Determine whether action is required 5. INGEST & ORCHESTRATE - Normalize, enrich, route and manage workflow 6. PERSIST STATE - Transaction state, context and audit 7. AI DECISION / REASONING - Contains three sub-components: - Retrieval → Policy → Prompt → LLM/AI - Reasoning → Structured Decision 8. GOVERN / DECIDE - Confidence, policy and human oversight 9.

[5]Enterprise_AI_Automation_Framework.docx0.3661

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The flow begins with "Enterprise Signals" at the top, which serves as the entry point for all data and events entering the system. The first stage, OBSERVE, captures signals and context from various enterprise sources. These signals progress to CORRELATE, where they are correlated, enriched, and analyzed to extract meaningful patterns and relationships. The third stage, QUALIFY / TRIGGER, evaluates the correlated data to determine whether any action is required, effectively filtering signals to identify relevant events requiring further processing. In stage four, INGEST & ORCHESTRATE, qualified signals are normalized, enriched, routed, and managed as workflows to ensure consistent processing across the enterprise. Stage five, PERSIST STATE, maintains transaction state, context, and audit information, ensuring traceability and governance throughout the process. The sixth stage, AI DECISION / REASONING, represents a critical juncture where the framework employs artificial intelligence and machine learning. This stage incorporates a detailed sub-process chain: Retrieval

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